

IIRS Outreach Program

AI/ML for Geodata Analytics

Schedule & Lecture Details

August 03 – 14, 2026

Lecture No.	Date/Time	Topic	Syllabus	Faculty
L1	August 3, 2026 1600 hrs	GIS Data analytics	Introduction to Data Analytics, Data Collection methods, Data Cleaning, Exploratory Data Analysis, Data Visualization tools, Statistics for Data Analysis	Dr. Harish Chandra Karnatak
L2	August 4, 2026 1600 hrs	Image Processing Methods	Introduction to Digital Images, Image Enhancement and Restoration, Image Transforms, Image segmentation, Object Detection	Dr. Poonam Seth Tiwari
L3	August 5, 2026 1600 hrs	Vector Data Processing	Introduction to Vector Data, Source of Vector Data, Vector Data Analysis- Buffer, Overlay etc.	Dr. Kamal Pandey
L4	August 6, 2026 1600 hrs	Concept of Data Science	Overview of Spatial Data Science: Approaches and Tools	Shri. Pankaj Bodani, SAC
L5	August 7, 2026 1600 hrs	Python for Image Processing	Basic Image Operations, Image Enhancement (Brightness and contrast adjustment, histogram equalization) , Geometric Transformations (Scaling, translation, rotation, affine transformations), Image Filtering (Smoothing, sharpening, edge detection (Sobel, Canny, Laplacian)	Mr. Ravi Bhandari
L6	August 10, 2026 1600 hrs	Machine Learning for geospatial analysis	Basics of Machine Learning/ Implementing Machine Learning algorithm for geospatial data processing	Mr. Ravi Bhandari
L7	August 11, 2026	ANN and Deep Learning	Introduction to AI in Geospatial Analysis, Introduction to Artificial	Dr. Poonam Seth Tiwari

	1600 hrs		Neural Networks (ANNs), Deep Learning Architectures for Geospatial Analysis: CNN, RNN, Geospatial Data Analysis Tasks: Image Classification, Object Detection and Segmentation (Instance and Semantic)	
L8	August 12, 2026 1600 hrs	AI/ML for Agriculture Analytics	Case Examples for AI/ML for Agriculture Analytics	Dr. Anil Kumar
L9	August 13, 2026 1600 hrs	Advances in AI/ML for Geo-data processing	GeoAI for Smart Governance: From Spatial Data to Intelligent Decision	Shri. Nilay Nilesch, NESAC
L10	August 14, 2026 1600 hrs	Generative AI and NLP for Geodata Analytics	Fundamentals of Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP) for Geodata, AI-powered Geospatial Analysis	Mr. Shubhangam Garg